Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 2

Team No.: 11

Project Title: Argument Passing

Team Members

Roll Number	Student Name	Signature
2520030141	Hima Varshini	
2520030142	Siri Chandana	
2520030442	Sai Pranathi	

1. Abstract

Argument Passing is a basic Operating System and systems programming concept that allows a program to receive information when it is started from the Linux command line. In C, command-line arguments are passed to the main() function through argc and argv. This project demonstrates how Linux programs receive, access, display, and process command-line arguments. The system will accept multiple arguments from the terminal, identify the number of arguments, display each argument, and perform simple operations such as searching, counting, or combining arguments. The project provides practical understanding of program execution, command-line interfaces, argument handling, and basic Linux process concepts. The solution will be implemented using C on Linux/Ubuntu and compiled using GCC. The project is simple to implement, easy to demonstrate, and useful for understanding fundamental Operating Systems and Systems Programming concepts.

2. Problem Statement

Linux programs often need to receive input at the time they are executed. Entering every input only after a program starts is not always convenient. Argument Passing provides a mechanism to pass information directly to a program through the command line. The project aims to develop a simple Linux-based C program that accepts command-line arguments, accesses them using argc and argv, displays them, and processes them correctly. This is relevant to Operating Systems and Systems Programming because it demonstrates how a program receives information during execution and interacts with the Linux command-line environment.

3. Objectives

1. To understand the concept of command-line argument passing in Linux.

2. To learn how argc and argv work in a C program.

3. To pass multiple pieces of information to a program during execution.

4. To display and process the received arguments.

5. To understand the relationship between Linux program execution and command-line input.

6. To develop a simple and user-friendly command-line application.

4. Proposed Methodology

1. Create a C program that accepts command-line arguments.

2. Use argc to determine the number of arguments supplied to the program.

3. Use argv[] to access each individual argument.

4. Display the arguments received from the Linux terminal.

5. Perform simple operations such as searching, counting, or combining arguments.

6. Compile the program using the GCC compiler.

7. Execute the program from the Linux terminal with different sets of arguments.

8. Verify that the arguments are correctly received and processed.

Example command:

./argument_demo Hello Linux OS

Expected example output:

Number of arguments: 4

Argument 0: ./argument_demo
Argument 1: Hello
Argument 2: Linux
Argument 3: OS

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project

Command-line arguments	Provides input to the program when it is launched from the Linux terminal.
argc	Stores the number of command-line arguments received by the program.
argv[]	Stores the command-line argument strings and provides access to each argument.
main()	Acts as the entry point of the C program and receives argc and argv.
strcmp()	Can be used to compare command-line arguments for simple processing.
exec() family (optional)	Can be used to demonstrate passing arguments while executing another program.
printf()	Displays arguments, messages, and processing results.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030141	Hima Varshini	Design and implement the Priority Scheduling algorithm
2520030142	Siri Chandana	Calculate scheduling parameters and generate Gantt Chart
2520030442	Sai Pranathi	Testing, output analysis, and project documentation

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system and terminal environment for developing and executing the project.
C	Used to implement the argument passing program.
GCC Compiler	Used to compile the C source code.
Linux Terminal	Used to execute the program and provide command-line arguments.
VS Code / Text Editor	Used to write and edit the C source code.

8. Expected Outcome

The project is expected to successfully demonstrate how command-line arguments are passed to and processed by a C program in Linux. The user will be able to provide multiple arguments through the terminal, and the program will identify, display, and process them correctly. The project will provide a clear practical understanding of Linux program execution, command-line interfaces, argc, argv, and basic systems programming concepts.

9. GitHub Repository

GitHub Repository Link: _____

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____